

ConceptBase V6.1 Programmer's Manual

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Contents

Chapter 1

Server Interface

This chapter provides basic information necessary for communication with the ConceptBase server. It is possible that the CBserver and the clients 'live' on different machines because communication with the CBserver is realized through a message protocol using inter-process communication (IPC) based on standard Internet sockets. It is even possible (but not recommended :-)) to use a standard telnet program for the communication with a CBserver. The following chapter describes this protocol as it is necessary to know for a specialized client which wants to request services from the CBserver. Readers who intend only to use one of the programming interfaces for C, C++ or Java may skip this chapter, but it contains some useful basic information.

From a client's point of view the CBserver can be seen as an abstract data type exporting several parameterized operations. These operations comprise methods for storing/retrieving information into/from the KB, methods for establishing and closing the connection to a CBserver and methods for testing the KB. Since the client and CBserver are two different processes a client cannot directly call these methods like procedures but must access them using a message protocol. However, the use of one of the application programming interfaces (API) for C, C++ and Java simplifies the communication and interaction with the CBserver from the viewpoint of an application programmer.

This chapter is organized as follows: Section ?? describes the message protocol which is used to communicate with other processes. Section ?? describes the interface to the CBserver, i.e. the data structures and operations which the ConceptBase kernel offers and the message protocol which makes these operations accessible to other processes.

1.1 Message Format

As already mentioned, any client that wants to use the methods of a CBserver has to communicate with CBserver according to a message protocol. So called ipcmessages can be sent via IPC to the port reserved for this CBserver. The CBserver handles such a message and reports back an answer: the ipcanswer.

1.1.1 ipcmessages

ipcmessages (**sender**, **receiver**, **method**, **args**), where

sender is the identifier for the sender of the message,

receiver is the identifier for the receiver of the message (usually the CBserver itself, but could be any other client connected to CBserver as well),

method is one of the methods exported by the CBserver (or a method known to another client which is addressed by the message),

args are the arguments for method.

Note, that it is necessary to “encode” the parameters of an ipcmessage. This means, that the strings must begin and end with ". If the string contains the characters " or \, they must be escaped with a backslash (\). Please refer to the grammar definition in appendix ?? for full details.

Messages can also be directed to other clients of the CBserver by using a different ID than the server ID as a receiver of message. If messages are sent from client to client, clients have to poll for messages using the method NEXT_MESSAGE. This function has not been tested recently.

A message can be prefixed by the length of the message, which is specified in five bytes. The first byte is always the character 'X', the next bytes are computed by the following formulas (len is the length of the message without this prefix):

1. $(len / 256^3) \text{ modulo } 256$
2. $(len / 256^2) \text{ modulo } 256$
3. $(len / 256) \text{ modulo } 256$
4. $len \text{ modulo } 256$

e.g., the first byte is the highest byte and the last byte is the lowest byte of an unsigned integer. Note, that specifying the length of an IPC-message is optional. IPC-messages without the length information should also be accepted by the server but communication problems might occur in rare circumstances.

1.1.2 ipcanswer

ipcanswer (**sender**, **completion**, **return**). where

sender is the identifier of the answering program (usually the CBserver since other programs cannot answer directly but only receive the message and send back another message via the CBserver). This is sent as an encoded string.

completion signals success (=ok) or failure (=error) or inability (=not_handled) of handling the message

return contains the return value(s) of the handled message. This is sent as an encoded string.

Additionally the CBserver administrates message queues for all connected clients. Whenever a client X sends a message to another client Y (which is not CBserver) the CBserver stores this message into the message queue of the client Y and gives it back to X.

Remark: If clients are likely to exchange messages they should periodically poll their message queue.

1.2 Methods Exported by the CBserver

The CBserver offers the following methods (list is incomplete):

general methods: TELL, UNTELL, TELL_MODEL, ASK, HYPO_ASK, NEXT_MESSAGE, ENROLL_ME, CANCEL_ME, GET_MODULE_CONTEXT

privileged methods: STOP_SERVER, REPORT_CLIENTS

internal methods: LPI_CALL

Privileged methods affect other clients connected to CBserver as well and should only be executed by an authorized client. That means, that only the *owner* of the ConceptBase server process may execute this methods.

The internal method LPI_CALL gives a client the possibility to call internal procedures of the ConceptBase server. This method is mainly useful for ConceptBase developers for debugging and analysing.

In the following description of the methods **return** refers to the respective parameter of **ipcanswer** (**sender**, **completion**, **return**).

For each error occurring during the execution of a method an error message is stored by the CBserver **receiver** in the message queue of the client **sender**. This error message can be fetched via a call of method NEXT_MESSAGE, see below.

1.2.1 TELL

ipcmessage (sender, receiver, TELL, [objects])

objects encoded string containing object descriptions in Telos represented as frames

return "yes" in case of success, "no" otherwise

The CBserver receiver checks the syntax of **objects** creating a parse tree for each object description, called *SMLfragment*. If no syntax error occurs the SMLfragments are transformed into an internal network representation with specialized rules and constraints. Those facts which are not already retrievable are temporarily added to the KB. A check is then performed to determine whether the updated KB still satisfies the integrity constraints. In the case of satisfaction the new information is made permanent, otherwise it is deleted.

1.2.2 UNTELL

ipcmessage (sender, receiver, UNTELL, [objects])

objects encoded string containing object descriptions in Telos represented as frames

return "yes" in case of success, "no" otherwise

The **objects** will be untold, i.e. the upper bound of their transaction time interval is set to the time the UNTELL operation takes place. That means from this time on the system does not believe this information anymore. Questions about the current state of the knowledge base yield the same answer as if the objects were never inserted into the system. However questions about earlier states will regard all information (even untold) the transaction time of which contains the time in question (= rollback time). Like in the TELL method, if the UNTELL operation would result in an inconsistent KB state it is rejected by the integrity checker.

1.2.3 TELL_MODEL

ipcmessage (sender, receiver, TELL_MODEL, [[fileList]]).

filelist A list of comma-separated ipc strings, which contain the full filenames of files to be loaded by ConceptBase server.

This method is similar to the TELL method, except that the frames which are told to ConceptBase are loaded from the given files and not passed directly to ConceptBase.

Remark: The files to be loaded by ConceptBase must be accessible for the server. This is not always the case, when server and client are running on different machines with different filesystems mounted on. Another problem may occur, due to access protections, because the user running the ConceptBase server is not allowed to read the specified files.

1.2.4 ASK

ipcmessage (sender, receiver, ASK, [Format, Query, AnswerRep, RollbackTime])

Format is either FRAMES or OBJNAMES, depending of the format of *Query*. If in *Query* only the object name of a query is given (e.g. AllEmployees) then the format must be OBJNAMES. If the query is specified as frame (e.g. "QueryClass AllEmployees isA Employee end") then the format must be FRAMES.

Query depending on the *Format* this may be simple object names or frames representing queries. In the later case, the query is temporarily told to the object base and after evaluating deleted from the object base, if it does not already exist in the object base before the transaction.

AnswerRep answer format specification, possible values are: FRAGMENT, FRAME, LABEL or an instance of *AnswerFormat*¹. The syntax of the FRAGMENT and FRAME formats

¹See the *ConceptBase User Manual* for details about user-defined answer formats

are explained in the appendix of the *ConceptBase User Manual*. If the answer representation is LABEL a comma-separated list of object names is returned.

RollbackTime rollback time specification

return list of answers in case of success, "no" otherwise

The values of the `Format` argument (FRAMES and OBJNAMES) are ipc message keywords and must not be encoded as the other arguments `Query`, `AnswerRep` and `RollbackTime`.

Example:

The following two queries are predefined builtin queries and available after booting the *ConceptBase* server. These queries additionally give good examples for derived expressions by instantiating parameters of generic query classes.

- `exists[x/objname]`
The answer return is "yes" if there is an object named x, otherwise "no".
- `get_object[x/objname]`
The answer is the frame representing the object x if there is an object x. Otherwise, the answer is "no". Only information that is explicitly stored (i.e. not inherited or deduced) is considered. If you want deduced information, you must specify additional parameters. For example, the answer of the following query is the `Class` object with stored and deduced attributes:

```
get_object[Class/objname,FALSE/dedIn,FALSE/dedIsa,TRUE/dedWith]
```

1.2.5 HYPO_ASK

`ipcmmessage(sender, receiver, HYPO_ASK, [ObjList, Format, Query, AnswerRep, RollbackTime])`

ObjList string of objects in frame syntax

Format see ASK

Query see ASK

AnswerRep see ASK

RollbackTime see ASK

return list of answers in case of success, no otherwise

This method allows to process so called 'hypothetical' queries against the KB. The objects in `objList` are temporarily told. This list may contain query objects which may in turn be referred to by names contained in `Query`. Then the queries in `queryList` are evaluated as if the temporary information would belong to the KB. Afterwards the temporary information will be removed.

1.2.6 NEXT_MESSAGE

`ipcmmessage ( sender, receiver, NEXT_MESSAGE, [type] )`

type identifier describing the type of the message (e.g. `ERROR_REPORT`). This argument may not be encoded as other string, but may be *empty*.

return contains the next message for the client if its message queue contains at least one message, `empty_queue` if no message exists

Client sender requests a message from the CBserver receiver stored in its message queue. Usually, this method is called after the CBserver returns error for a previous method. The client program must then get all error messages until it gets "`empty_queue`" as answer.

1.2.7 STOP_SERVER

ipcmessage (sender, receiver, STOP_SERVER, [password])

password password allowing a client to stop a CBserver (may be *empty*)

return "yes" in case of success, "no" otherwise

The CBserver receiver is terminated if the password is correct and the user running the client is also the owner of the CBserver to be stopped. To the requesting client `STOP_SERVER` has the same effect as `CANCEL_ME`. It is recommended to terminate the CBserver by using the respective menu choice from the "Server Menu" of ConceptBase Workbench if you want to stop the CBserver process.

1.2.8 REPORT_CLIENTS

ipcmessage (sender, receiver, REPORT_CLIENTS, [])

return list of all clients currently connected to CBserver receiver

CBserver receiver reports back the identifier, toolclass and owner name of all currently connected clients including itself.

1.2.9 ENROLL_ME

ipcmessage (sender, receiver, ENROLL_ME, [toolclass,username])

toolclass 'class' the client belongs to

username name of the user running the client

return identifier assigned to the client by CBserver

`sender` and `receiver` have value "" since they are not known. The sending client will be registered as a new client of the CBserver with its own identifier and message queue. This message must be sent to the CBserver before any other message can be sent, since all other messages require valid identifiers to be assigned to sender and receiver. If the user specified by `username` is stored as an instance of the class `CB_User` of the CBserver, then the value of the attribute `homeModule` of that user is taken as the initial module context of the client. Otherwise, the default module context `System` is assigned to the client. In a variant of `ENROLL_ME`, one can specify a third parameter `module` which will set the module context explicitly.

1.2.10 CANCEL_ME

ipcmessage (sender, receiver, CANCEL_ME, [])

return "yes" in case of successful disconnection, "no" otherwise

Client sender will be disconnected from CBserver. This means that from now on the sender is no longer known to the CBserver (no further messages can be sent) and its message queue is deleted. After successfully canceling the connection, the ipc sockets to the server must be closed by the client program.

1.2.11 GET_MODULE_CONTEXT

ipcmessage (sender, receiver, GET_MODULE_CONTEXT, [])

return name of the module currently assigned to client `sender`

This service allows clients to interrogate the CBserver about the currently active module context in which they operate.

1.2.12 LPI_CALL

ipcmesssage (sender, receiver, LPI_CALL, [call])

call an internal routine

return "yes" if call succeeded, "no" otherwise

This is for debugging and testing purposes only.

Chapter 2

Programming Interface for a C Client: libCB

This chapter describes the programming interface for ConceptBase. The programming interface consists of a number of data structures and C functions which are defined in the header file `CBinterface.h`. Make sure that this header file is included in each of source files that use functions of `libCB`. The data structures are explained in section ???. The C library `libCB` contains all functions described in section ???.

The libraries can be found in the following directories:

Solaris/SPARC: The directory `$CB_HOME/sun4/lib` contains the libraries for static linking of your application.

Solaris/PC: The directory `$CB_HOME/i86pc/lib` contains the libraries for static linking of your application.

Linux: The directory `$CB_HOME/linux/lib` contains the libraries for static linking of your application.

Windows: The directory `$CB_HOME/windows/lib` contains the dynamic libraries for dynamic linking of your application.

There are currently no plans to build dynamic libraries for the Unix-based platforms.

The directories `$CB_HOME/examples/Clients/LogClient` and `$CB_HOME/examples/Clients/C_Client` contain example programs, which uses the programming interface `libCB` to communicate with the ConceptBase server. The `LogClient` program is explained in appendix ???. Information on you how to compile and link your source code with the ConceptBase libraries can be found either in the directories of the example clients or in section ???.

2.1 Data Structures

This section describes the data structures used by the API, in particular structures which are passed to and returned by the interface procedures.

The following C-types are defined in the file `CBinterface.h` which is located in the directory `$CB_HOME/include`.

2.1.1 Completion

```
typedef enum {CB_OK=0, CB_ERROR, CB_NOT_HANDLED,  
              CB_TIMEOUT, CB_CONN_BROKEN} Completion;
```

The different return values have to be interpreted as follows:

CB_OK the message has been handled successfully.

CB_ERROR an error occurred during the execution of the message; the ConceptBase server stores some error reports for you on your message queue which may be read calling `em = get_errormessages()` (see below).

CB_NOTIFICATION indicates that the message is a notification message. Notification messages are sent by the server if the client has requested notification on updates on certain views.

CB_NOT_HANDLED the server was not able to manage your message at all. This may be due to an invalid format of input parameters (e.g. wrong Telos syntax) or missing parameters.

CB_TIMEOUT the message has been sent successfully to the server, but there has been no answer from the server after a specific amount of time (depends on the type of message sent). This may be due to the number of clients which are active or due to the kind of message you sent (some queries may last longer than others). The client is responsible for the correct handling of answers returned after **CB_TIMEOUT** occurred.

CB_CONN_BROKEN the sending of the last message failed (the connection to the server is no longer accessible). Again, the client is responsible to handle this return value (e.g. stopping the client).

2.1.2 Answer

```
struct answer { char *sender;
                Completion completion;
                char *return_data;
            };
typedef struct answer Answer;
```

The `Answer` structure is returned by most library functions. The first field `sender` contains the name of the sender as it is maintained by the ConceptBase server. The second one specifies the status of the message processing (see section ??) while the third one contains return values of the message called.

2.1.3 Server

```
struct server { char *serverName;
                char *client;
                int connected_to_CB_server;
                SOCKET socket;
            };
typedef struct server Server;
```

This structure is allocated and filled by the `connect_CB_server()` call and used as an anchor by all the other routines to get the right server. The field `connected_to_CB_server` should usually be true, as it indicates that the client is connected to the server (or not). The `socket` field represents the socket which is used for the communication with the CB server and should be used only internally.

2.1.4 Clients

```
struct clients { char *client;
                 char *toolclass;
                 char *username;
                 struct clients *next;
            };
typedef struct clients Clients;
```

This structure represents a simply linked list of clients. A pointer to this structure is returned as result of the `report_clients` call.

2.1.5 Error_Messages

```
struct errormessages { char *errormessage;  
                      struct errormessages *next;  
                      };  
typedef struct errormessages Error_Messages;
```

List of error messages given by the ConceptBase server every time a communication event can not be processed correctly. This list may be obtained calling `get_errormessages()`.

2.2 Functions

2.2.1 connect_CB_server

```
int connect_CB_server(int    portnr,  
                     char    *hostname,  
                     char    *clientname,  
                     char    *username,  
                     Server **server)
```

Description:

Sets up a connection to a given ConceptBase server. This routine has to be called once before calling one of the following routines.

Input parameters:

portnumber number of the port of the server (this port number is unique per server as may be defined at the server's start up time).
hostname name of the machine on which you started the server
clientname name of the client to be connected (e.g. *TelosEditor*)
username name of the user who started the client
server pointerpointer to a struct server; on a succesfull connection the structure will be allocated and filled

Result:

0 Connection established
-1 There is no such server (probably wrong portnumber and/or host)
>0 a completion value (see section ??)

2.2.2 disconnect_CB_server

```
int disconnect_CB_server(Server *server)
```

Description:

Closes a previous connection to a ConceptBase server. This procedure has to be called every time a client is stopped (but usually the CBserver is not affected by clients that crash or do not disconnect correctly).

Input parameters:

server pointer to the structure discribing the current ConceptBase server

Result:

0 Connection correctly terminated
-1 error, not connected
>0 a Completion value

2.2.3 tellCB

```
Answer* tellCB(Server *server, char *objects)
```

Description:

Inserts a set of objects into the ConceptBase server. This function has been renamed from previous releases as `tell` is a operating system function on some systems.

Input parameters:

server pointer to the structure describing the actual server

objects pointer to a list of objects, which should be inserted into the knowledge base. This should be a normal NULL-terminated C-string.

Result:

An answer struct where `return_data` is either `yes` or `no` and where the completion value indicates the result of the operation:

CB.OK operation successful

CB.ERROR There was an error while inserting, get the error messages by calling `get_errormessages()`

other see the description in section ??

2.2.4 untell

```
Answer* untell(Server *server, char *objects)
```

Description:

Removes a list of objects from the knowledge base. Note the specific semantics of the `untell` method as described in chapter ?? of this Manual.

Input parameters:

server pointer to the structure describing the actual server

objects pointer to a list of objects, which should be deleted. This should be a normal NULL-terminated C-string.

Result:

An answer struct where `return_data` is either `yes` or `no` and where the completion value indicates the result of the operation:

CB.OK operation successful

CB.ERROR There was an error while removing, get the error messages calling `get_errormessages()`

other see the description in section ??

2.2.5 tell_model

```
Answer* tell_model(Server* server, char** models);
```

Description:

Tells the given files to the server. Note that the server must be able to find these files in its file system.

Input parameters:

server pointer to the structure describing the actual server

objects pointer to a NULL-terminated array of C-strings, containing the file names which should be loaded by the server.

Result:

An answer struct where `return_data` is either yes or no and where the completion value indicates the result of the operation:

CB.OK operation successful

CB.ERROR There was an error while removing, get the error messages calling `get_errormessages()`

other see the description in section ??

2.2.6 get_errormessages

```
Error_Messages *get_errormessages(Server *server)
```

Description:

Gets the error messages corresponding to the last error. This procedure has to be called every time **CB.ERROR** has been returned by a given procedure. Otherwise, further messages may be disturbed by the error messages which are returned first by the server.

Input parameters:

server pointer to the structure describing the actual server

Result:

list of error messages (see section ??)

2.2.7 ask

```
Answer* ask(Server* pServer,  
            char* szQuery,  
            char* szAskFormat,  
            char* szAnsFormat,  
            char* szRBTime);
```

Description:

Sends the query in the specified format (`szAskFormat`) to the server and returns the result of the server, which will be represented in the format given in `szAnsFormat`. The rollback time (`szRBTime`) is usually Now.

Input parameters:

pServer a pointer to a server structure

szQuery the query

szAskFormat the format of the query (FRAMES or OBJNAMES)

szAnsFormat the format of the answer (e.g. FRAME, LABEL,...)

szRBTime rollback time (e.g. Now)

Result:

an answer struct:

sender the tool that has provided the answer, usually the ID of the server

completion Completion value indicating the success of the method, e.g. **CB.OK**, **CB.ERROR**

return_data the result of the query in the specified format, or the string "nil" if there are no results or if there was an error during query processing

2.2.8 ask_frames

```
Answer* ask_frames(Server *pSserver,  
                   char *szQuery,  
                   char* szAnsFormat,  
                   char *szRBTime)
```

Description:

As ask, but szAskFormat is fixed to be FRAMES, i.e. queries have to be given as frames.

2.2.9 ask_objnames

```
Answer* ask_objnames(Server *pSserver,  
                     char *szQuery,  
                     char* szAnsFormat,  
                     char *cbfoGet,  
                     char *szRBTime)
```

Description:

As ask, but szAskFormat is fixed to be OBJNAMES, i.e. queries have to be given as object names (or derive expressions).

2.2.10 hypo_ask

```
Answer* hypo_ask(Server* pServer,  
                  char* szFrames,  
                  char* szQuery,  
                  char* szAskFormat,  
                  char* szAnsFormat,  
                  char* szRBTime);
```

Description:

As ask, but first tells the frames given in szFrames to the server, then performs the query and finally deletes the told frames from the object base.

2.2.11 report_clients

```
Clients* report_clients(Server *server)
```

Description:

Returns a list of all clients connected to the server.

Input parameters:

server pointer to the structure describing the actual server

Result:

list of clients or NULL on error

2.2.12 get_servermessage

```
Answer* get_servermessage(Server* server, char* type);
```

Description:

Gets a message from the server for the client. This function is called by `get_errormessages()`.

Input parameters:

server a pointer to a server structure

type type of the message to be retrieved (e.g. ERROR_REPORT)

Result:

an answer object with the message or EMPTY_QUEUE in `return_data`

2.2.13 get_notification

```
Answer* get_notification(Server* server, int timeout);
```

Description:

Looks for a notification message. Notification messages are sent by the server if the client has requested notification on updates on certain views. The method will wait for a message from the server for the specified time.

Input parameters:

server a pointer to a server structure

timeout time to wait for a message

Result:

an answer object with completion CB_NOTIFICATION when a message was received, otherwise a completion value, usually CB_TIMEOUT.

2.2.14 stopServer

```
Answer* stopServer(Server* server, char* password);
```

Description:

Stops the server. Note, that only the user who has started the server may stop it.

Input parameters:

server a pointer to a server structure

password a password (not used, may be empty)

Result:

the result of the method

2.2.15 LPICall

```
Answer* LPICall(Server* server, char* lpicall);
```

Description:

Performs a LPI-Call at the server. With LPI (Logic Programming Interface) one can call ProLog predicates defined in an LPI-Module.

Input parameters:

server a pointer to a server structure
lpicall the predicate to be called

Result:

the result of the method

2.2.16 free*

```
void freeAnswer(Answer* ans);  
void freeServer(Server* srv);  
void freeClients(Clients* c);  
void freeErrorMessages(Error_Messages* err);
```

Description:

These functions free the allocated memory by the corresponding structures. Note that memory of all results which are returned by the library methods have to be freed by the caller.

2.2.17 send_message

```
Answer send_message(Server *server,  
                    char *method,  
                    char *data)
```

Description:

This procedure is the most general one and used by most functions mentioned before. It sends a message of type `method` to the (already connected) CBserver `server`. `data` is a string containing data expected by the method `method`¹. For normal usage of the client library, this function is not necessary. The more specific functions (e.g. `tellCB`, `untell`, ...) are more useful.

Input parameters:

server pointer to the structure describing the actual server
method string which defines the type of the message (e.g. `TELL`)
data the arguments for the given message type (e.g. `["Class Employee with ... end"]`)

Result:

an Answer structure containing sender, completion and return_data

¹See chapter ?? and appendix ?? of this manual for a complete description of the available methods and their expected data

2.2.18 CBdecodeString

```
char* CBdecodeString(const char* s);
```

Description:

Decode a string. ConceptBase encodes all strings with "" and . To get the plain string, use this function.

Input parameters:

The string to decode.

Result:

The decoded string, it is a duplicate of the input if the input string is not encoded. The memory allocated by the result has to be freed by the caller.

2.2.19 CBencodeString

```
char* CBencodeString(const char* s);
```

Description:

Encode a String. ConceptBase encodes all strings with "" and . Use this function if you want to use Strings in Telos frames.

Input parameters:

The string to encode.

Result:

The encoded string. The memory allocated by the result has to be freed by the caller.

2.2.20 CBgetEncodedLength

```
unsigned CBgetEncodedLength(const char* s);
```

Description:

Return the length of an encoded string. This function is called by CBencodeString to allocate the memory of the encoded string.

2.2.21 CBgetLabels

```
char** CBgetLabels(const char* labelList);
```

Description:

Parse a comma-separated list of labels. ConceptBase returns sometimes comma-separated list of labels (e.g., for the answer format LABEL). This function makes an array of strings out of one plain string. This is a lazy function that will fail to produce a correct result if the object names contain commata (e.g., "This, is, a, Telos, object, name, with, commata.>").

Input parameters:

A string with comma-separated-list.

Result:

A NULL-terminated array of strings.

2.3 Compiling and Linking

If you want to *compile* your source that uses libCB, you have basically to make sure two things:

- the header files of ConceptBase are found, and
- the correct system header files are included in `CBinterface.h`

The first item is usually achieved by adding a parameter `-I` with the include-directory of your ConceptBase installation to the list of compiler options. For the second point, you have to define the symbol `LINUX`, `WIN32` or `SOLARIS` (usually done with the `-D` option of the compiler), depending on the operating system of your client application.

We have used the following compiler flags (with gcc 3.2 on the UNIX-based systems, and MS Visual C++ 6.0 on Windows):

Solaris `-I$(CB_HOME)/include -DSOLARIS`

Linux `-I$(CB_HOME)/include -DLINUX`

Windows `-nologo -MT -W3 -GX -O2 -I$(CB_HOME)/include -D "WIN32"
-D "NDEBUG" -D "_CONSOLE" -D "_MBCS" -Fo".\\" /Fd".\\" -c`

If you want to *link* your application, you have to make sure that libraries are found by the system (`-L` option of gcc) and that the library libCB is indeed linked to your application (`-l` option). We have used the following linker options:

Solaris `-L$(CB_HOME)/sun4/lib -lCB -lnsl -lsocket`

Linux `-L$(CB_HOME)/linux/lib -lCB`

Windows `kernel32.lib user32.lib wsock32.lib $(CB_HOME)/windows/lib/libCB.lib
-nologo -subsystem:console -incremental:no -machine:I386`

Chapter 3

Programming Interface for a C++ Client: libCBview

The libCBview provides a C++ encapsulation of libCB. It provides only an object-oriented API for ConceptBase and does not provide any additional methods in contrast to libCB.

Compilation and linking has to be done in the same way as for libCB. Note that you have to link both libraries libCB and libCBview if you want to use the C++ classes.

The documentation in the following sections has been generated with DOC++ (<http://docpp.sourceforge.net>).

3.1 CBclient

class CBclient

A client class for ConceptBase

3.1.1 CBclient

CBclient ()

Description:

Constructs an "empty" client which is not connected

3.1.2 CBclient

CBclient (char* host, int port, char* tool=(char*)NULL, char* user=(char*)NULL)

Description:

Constructs a new CBclient object and connect to the specified host

3.1.3 ~CBclient

virtual ~CBclient ()

Description:

Disconnects from the CBserver and deallocates the memory

3.1.4 tell

CBAnswer* tell (char*)

Parameters:

char *frames the frames

Returns:

- a CBAnswer object containing the result and the completion

Description:

Tells frames to the server

3.1.5 untell

CBAnswer* untell (char*)

Parameters:

char *frames the frames

Returns:

- a CBAnswer object containing the result and the completion

Description:

Untells frames to the server

3.1.6 tellModel

CBAnswer* tellModel (char)**

Parameters:

char** files an array of filenames

Returns:

- a CBAnswer object containing the result and the completion

Description:

Tells files containing frames to the server

3.1.7 ask

CBAnswer* ask (char* query, char* format="OBJNAMES", char* answerrep="FRAME", char* rollbacktime="Now")

Parameters:

char *query the query

char* format the format of the query (FRAMES or OBJNAMES)

char* answerrep the format of the answer (FRAME)

char* rollbacktime Rollback Time (e.g. "Now")

Returns:

- a CBAnswer object containing the result and the completion

Description:

Sends a query to the ConceptBase server

3.1.8 hypoAsk

CBanswer* hypoAsk (char* frames, char* query, char* format="OBJNAMES", char* answerrep="FRAME", char* rollbacktime="Now")

Parameters:

char *frames frames to be told
char *query the query
char* format the format of the query (FRAMES or OBJNAMES)
char* answerrep the format of the answer (FRAME)
char* rollbacktime Rollback Time (e.g. "Now")

Returns:

- a CBanswer object containing the result and the completion

Description:

Sends frames and a query to the ConceptBase server. The frames are told temporarily, the query is evaluated, and the temporarily objects are removed.

3.1.9 askObjNames

CBanswer* askObjNames (char* query, char* answerrep="FRAME", char* rollbacktime="Now")

Parameters:

char *query the query
char* answerrep the format of the answer (FRAME)
char* rollbacktime Rollback Time (e.g. "Now")

Returns:

- a CBanswer object containing the result and the completion

Description:

Sends a query to the ConceptBase server. Same as ask but with fixed query format (OBJNAMES).

3.1.10 askFrames

CBanswer* askFrames (char* query, char* answerrep="FRAME", char* rollbacktime="Now")

Parameters:

char *query the query
char* answerrep the format of the answer (FRAME)
char* rollbacktime Rollback Time (e.g. "Now")

Returns:

- a CBanswer object containing the result and the completion

Description:

Sends a query to the ConceptBase server. Same as ask but with fixed query format (FRAMES).

3.1.11 enrollMe

int enrollMe (char* host, int port, char* user=NULL, char* tool=NULL)

Parameters:

host hostname of the machine where the server runs
port port number of server
*user the name of the tool
*tool the name of the user

Description:

Connects to a ConceptBase Server Return the return value of connect_CB_server (see CBinterface.h): -1: if socket to specified can not be opened 0: ok other: a completion value (see CBinterface.h)

3.1.12 cancelMe

int cancelMe ()

Description:

Disconnects from a ConceptBase Server

Return the return value of disconnect_CB_server (see CBinterface.h): -1: error, not connected 0: ok other: a completion value (see CBinterface.h)

3.1.13 stopServer

CBAnswer* stopServer (char* password=NULL)

Returns:

- a CBAnswer object containing the result and the completion

Description:

Stops the ConceptBase server. Note that a server may be stopped only by the user who has started it.

3.1.14 reportClients

Clients* reportClients ()

Description:

Return a list of clients connected to the CB server. The result will be a list of Client objects as defined in libCB.

3.1.15 nextMessage

CBAnswer* nextMessage (char* method="")

Parameters:

char* method the type of the message to be retrieved

Returns:

- a CBAnswer object containing the result and the completion

Description:

Gets a message from the server

3.1.16 `getErrorMessages`

CBError* `getErrorMessages` ()

Returns:

- a string containing all error messages

Description:

Gets the error messages from the server

3.1.17 `LPICall`

CBanswer* `LPICall` (char*)

Description:

Perform a LPI call on the server. A LPI call is a call of Prolog-predicate of the CBserver. This is mostly used for debugging.

3.1.18 `connected`

inline int `connected` ()

Description:

Check whether this client is connected

3.1.19 `operator int`

inline operator int ()

Description:

The operator int checks also if the client is connected.

3.1.20 `getServerName`

char* `getServerName` ()

Description:

Return the name of the server

3.1.21 `getClientName`

char* `getClientName` ()

Description:

Return the name of the client

3.2 `CBanswer`

class `CBanswer`

C++ Wrapper for Answer struct of libCB

3.2.1 CBanswer

CBanswer (Answer* ans)

Parameters:

ans pointer to the Answer struct

Description:

Constructs a CBanswer object from a Answer struct

3.2.2 ~CBanswer

~CBanswer ()

Description:

Deallocate the memory of the object

3.2.3 getCompletion

Completion getCompletion ()

Description:

Get the completion value of the answer

3.2.4 getResult

char* getResult ()

Description:

Get the result string of the answer

3.2.5 getRespondingTool

char* getRespondingTool ()

Description:

Get the ID of the responding tool of the answer. This usually the CBserver

3.3 CError

class CError

C++ Wrapper of Error_Messages struct in libCB.

3.3.1 CError

CError (Error_Messages* e)

Parameters:

e pointer to the Error_Messages

Description:

Construct a CError object from a list of Error_Messages

3.3.2 ~CError

~CError ()

Description:

Deallocate the memory of a CError object

3.3.3 getErrorMessage

char* getErrorMessage ()

Description:

Get the error message of this object. This will return only the first error message of the list.

3.3.4 getAllErrorMessages

char* getAllErrorMessages ()

Description:

This method will return all error messages of the list. The method will allocate a new string, thus the resulting string has to be freed by the caller.

3.3.5 getNextError

CError* getNextError ()

Description:

Get the next error message in the list

Chapter 4

Processing of Telos Frames: libtelos

This chapter explains the library `libtelos`, which contains the Telos parser. The Telos parser is able to parse the answers in FRAME or LABEL format from ConceptBase.

To call the Telos parser, you must link your program with the library `libtelos.a/libtelos.dll` which can be found in the directory `$CB_HOME/<arch>/lib` where `<arch>` is either `sun4`, `i86pc`, `linux`, or `windows`.

In your source files, you must include the header files `fragment.h`, `te_access.h`, `te_callparser.h`, `te_cursor.h`, and/or `te_smlutil.h`. All header files are located in the directory `$CB_HOME/include`.

The following sections explain several functions to call the parser and to handle the data structures. In principle, there are three different ways to parse and to access Telos frames:

- Using the functions and data structures defined in `fragment.h`, `te_callparser.h`, and `te_smlutil.h`: The Telos parser is invoked directly and the contents of the Telos frames is retrieved by navigating over a list of fragments (a fragment is a data structure for a Telos frame). See section ?? for details.
- Using the functions and data structures defined in `te_access.h`: Telos frames are represented in vectorized structure. One can use functions to create, destroy or apply to filters to the structure. See section ?? for details.
- Using the functions and data structures defined in `te_cursor.h`: Iterating over a set of Telos frames is done by using a cursor. See section ?? for details.

The documentation in the following sections has been generated with DOC++ (<http://docpp.sourceforge.net>).

4.1 fragment.h and te_callparser.h

4.1.1 Typedef: BindingList

typedef struct bindingList BindingList

Description:

A binding list represents the list of parameters in a derive expression

4.1.2 Typedef: ObjectIdentifier

typedef struct objectIdentifier ObjectIdentifier

Description:

An object identifier represents a Telos object name. It may be a simple object name, a derive expression, or a select expression.

4.1.3 Typedef: te_ClassList

typedef struct classlist te_ClassList

Description:

A class list is a list of object identifiers

4.1.4 Typedef: AttrClassList

typedef struct attrclasslist AttrClassList

Description:

An AttrClassList is a list of attribute categories. Attribute categories or simple labels.

4.1.5 Typedef: SpecObjId

typedef struct specObjId SpecObjId

Description:

Used only internally for extended syntax

4.1.6 Typedef: SelectExpB

typedef struct selectexpb SelectExpB

Description:

Used only internally for extended syntax

4.1.7 Typedef: Restriction

typedef struct restriction Restriction

Description:

Used only internally for extended syntax

4.1.8 Typedef: ObjectSet

typedef struct objectset ObjectSet

Description:

Used only internally for extended syntax

4.1.9 Typedef: PropertyList

typedef struct propertylist PropertyList

Description:

A property list is a list of attributes. Attributes have a label and a value. The member objectSet is used only in an extended syntax.

4.1.10 Typedef: AttrDeclList

typedef struct attrdecllist AttrDeclList

Description:

An AttrDeclList is a list of attribute declarations. It represents everthing between "with" and "end" in a Telos frame. One attribute declaration has a list of attribute categories and a list of properties (attribute definitions).

4.1.11 Typedef: te_SMLfragmentList

typedef struct smlfragmentList te_SMLfragmentList

Description:

A SMLfragmentList is a list of Telos frames. Each Telos frame has an object identifier (id). It may have in addition an inOmega class, a list of in-Classes, a list of isA-Classes, and an attribute declaration. Except id, all members may be NULL.

4.1.12 Typedef: FrameParseOutput

typedef struct frameParseoutput FrameParseOutput

Description:

FrameParseOutput is the structure returned by the function `te_frame_parser`. It contains either a list of fragments or information about the parse error.

<code>te_SMLfragmentList* smlfrag</code>	<i>the list of fragments</i>
<code>int error</code>	<i>0 if ok, 1 if parse error, 2 if input is null</i>
<code>char* errortoken</code>	<i>If there was an parse error, this should indicate the token that caused the error.</i>
<code>int errorline</code>	<i>If there was an parse error, this should be the line number of the error.</i>

4.1.13 Typedef: ClassListParseOutput

typedef struct classlistParseoutput ClassListParseOutput

Description:

ClassListParseOutput is the structure returned by the function `te_classlist_parser`. It contains either a list of classes or information about the parse error.

<code>te_ClassList* classlist</code>	<i>A list of classes (object names)</i>
<code>int error</code>	<i>Non-zero if an error occured</i>
<code>char* errortoken</code>	<i>If there was an parse error, this should indicate the token that caused the error.</i>
<code>int errorline</code>	<i>If there was an parse error, this should be the line number of the error.</i>

4.1.14 `te_frame_parser`

FrameParseOutput* `te_frame_parser` (char* indata)

Parameters:

`indata` a string containing the input frames

Returns:

a pointer to a `FrameParseOut` structure

Description:

Calls the Telos Parser to parse frames.

4.1.15 `te_classlist_parser`

`ClassListParseOutput* te_classlist_parser (char* indata)`

Parameters:

`indata` a string containing the object names

Returns:

a pointer to a `ClassListParseOut` structure

Description:

Calls the Telos Parser to parse a list of object names.

4.1.16 `FragmentToString`

`char* FragmentToString (te_SMLfragmentList *cursor`

Description:

Unparse a fragment list into a string

4.1.17 `DestroySMLfrag`

`void DestroySMLfrag (te_SMLfragmentList* fragment)`

Description:

Destroy a fragment list

4.1.18 `Destroy_ClassList`

`void Destroy_ClassList (te_ClassList* clist)`

Description:

Destroy a class list

4.2 `te_access.h`

4.2.1 `Structure: te_AttrDecl`

`struct te_AttrDecl`

Description:

This structure represents an attribute declaration. An attribute declaration is a list of attribute categories with a list of properties (label and values) that belong to these attribute categories.

<code>char** aszCategory</code>	<i>contains the list of category labels</i>
<code>char** aszLabel</code>	<i>contains the list of property labels corresponding to</i>
<code>char** aszValue</code>	<i>the list of property values</i>

4.2.2 Typedef: TAttrDecl

```
typedef struct te_AttrDecl TAttrDecl
```

4.2.3 Typedef: PAttrDecl

```
typedef TAttrDecl* PAttrDecl
```

Description:

The pointer for TAttrDecl

4.2.4 Typedef: VTelos

```
typedef struct te_VectorizedTelosframe VTelos
```

Description:

The type for te_VectorizedTelosframe

4.2.5 Typedef: PVTelos

```
typedef VTelos* PVTelos
```

Description:

A pointer to VTelos

4.2.6 Typedef: AVTelos

```
typedef PVTelos* AVTelos
```

Description:

An array of VTelos pointers

4.2.7 Structure: te_TelosReport

```
struct te_TelosReport
```

Description:

A te_TelosReport is a projection on certain attributes of a frame

```
char** aszLabel
```

List of labels in the report

```
char** aszValue
```

List of values in the report

4.2.8 Typedef: TReport

```
typedef struct te_TelosReport TReport
```

4.2.9 Typedef: PReport

```
typedef TReport* PReport
```

4.2.10 vt_createByFragment

AVTelos vt_createByFragment (te_SMLfragmentList* fl)

Parameters:

fl contains the fragment list in the way produced by the parser

Returns:

NULL if the argument is NULL too, else the pointer to the vector tree structure

Description:

Maps the given fragmentlist fl into a vector of frames.

4.2.11 vt_create

AVTelos vt_create (char* szTelos)

Parameters:

szTelos should be a string of correct Telos

Returns:

NULL if the szTelos fails the parsing process else it contains the AVTelos with all its componends

Description:

Maps the given Telos text szTelos into a vector of frames.

4.2.12 vt_destroy

void vt_destroy (AVTelos avtFrames)

Parameters:

avtFrames points to the vector tree structure

Description:

Disposes the given vector tree.

4.2.13 rep_create

PReport rep_create (PVTelos pvtFrame, char** aszCategories)

Parameters:

pvtFrame points to a single Frame, which must exists@pararm
aszCategories should be a NULL terminated vector of the categories to filter as a conjunction.

Returns:

In each case a report will be created, even if the result is empty.

Description:

Filters all attributes to those properties which belong to all given categories at the same time.
Note: If there is a category wrong typed, it has the effect that result will always be an empty vector with NULL at index 0.

4.2.14 rep_destroy

void rep_destroy (PReport prepReport)

Parameters:

prepReport points to the report which should be disposed

Description:

Disposes the given report structure. Should be called to free the result of rep_create.

4.2.15 getValueOfLabel

char* getValueOfLabel (PVTelos pvtFrame, char* szLabel)

Parameters:

pvtFrame points to a single frame, which should be analyzed
szLabel contains the keyword, which should be searched in the labels

Returns:

the string containing the value according to the given label
NULL if no appropriate value was found

Description:

A simple service routine, which support the access on frames.

4.2.16 getCategories

char getCategories (PVTelos pvtFrame)**

Parameters:

pvtFrame points to a single frame, which should be analyzed

Returns:

array of strings with the categories

Description:

Lists the categories as flat list, each element appears only once. The categories will be ordered by their appearance.

4.2.17 destroyASZ

void destroyASZ (char asz)**

Parameters:

asz the array of strings to be disposed

Description:

Disposes an asz (array of strings) structure.

4.3 te_cursor.h

4.3.1 Structure: te_framecursor

struct te_framecursor

Description:

A cursor for a Telos frame

<code>te_SMLfragmentList* flAll</code>	<i>Parsed telos frames in a fragment list</i>
<code>te_SMLfragmentList* flCur</code>	<i>fragment list cursor</i>
<code>te_ClassList* clCurOmega</code>	<i>cursor for inOmega</i>
<code>te_ClassList* clCurIn</code>	<i>cursor for in</i>
<code>te_ClassList* clCurIsA</code>	<i>cursor for isA</i>
<code>AttrDeclList* alCur</code>	<i>cursor for attribute declarations</i>
<code>AttrClassList* clCurCategory</code>	<i>cursor for attribute categories (within one attribute declaration)</i>
<code>PropertyList* plCur</code>	<i>cursor for property list (within one attribute declaration)</i>
<code>AttrDeclList* alChecked</code>	<i>internal filter: tests if alCur was checked</i>

4.3.2 Typedef: TFrameCursor

typedef struct te_framecursor TFrameCursor

4.3.3 Typedef: PFrameCursor

typedef TFrameCursor* PFrameCursor

4.3.4 te_createCursor

PFrameCursor te_createCursor (te_SMLfragmentList* fl)

Description:

Creates and initializes a cursor structure for the given smlfragmentlist. and returns a pointer to it.

4.3.5 te_destroyCursor

void te_destroyCursor (PFrameCursor pfc)

Description:

Deallocates a cursor structure

4.3.6 te_resetFrame

void te_resetFrame (PFrameCursor pfc)

Description:

Sets the frame cursor to the first frame and resets all sub cursors

4.3.7 te_nextFrame

int te_nextFrame (PFrameCursor pfc)

Returns:

`true` (non-zero) if there is a next element

Description:

Sets the frame cursor to the next frame in the list and resets the Omega, IsA, In, AttrDecl, Category and Property cursors.

4.3.8 te_retOID

char* te_retOID (PFrameCursor pfc)

Description:

Returns the OID of a frame as plain string, even if it is a select expression. Memory for this string has to be deallocated by the caller.

4.3.9 te_resetOmega

void te_resetOmega (PFrameCursor pfc)

Description:

Resets the Omega cursor

4.3.10 te_nextOmega

int te_nextOmega (PFrameCursor pfc)

Returns:

`true` (non-zero) if there is a next element

Description:

Sets the omega cursor to the next element.

4.3.11 te_retOmega

char* te_retOmega (PFrameCursor pfc)

Description:

Returns the omega object as string

4.3.12 te_resetIsA

void te_resetIsA (PFrameCursor pfc)

Description:

Resets the Isa cursor

4.3.13 te_nextIsA

int te_nextIsA (PFrameCursor pfc)

Returns:

`true` (non-zero) if there is a next element

Description:

Sets the IsA cursor to the next element.

4.3.14 te_retIsA

char* te_retIsA (PFrameCursor pfc)

Description:

Returns the IsA object as string

4.3.15 te_resetIn

void te_resetIn (PFrameCursor pfc)

Description:

Resets the In cursor

4.3.16 te_nextIn

int te_nextIn (PFrameCursor pfc)

Returns:

`true` (non-zero) if there is a next element

Description:

Sets the In cursor to the next element.

4.3.17 te_retIn

char* te_retIn (PFrameCursor pfc)

Description:

Returns the In object as string

4.3.18 te_resetAttrDecl

void te_resetAttrDecl (PFrameCursor pfc)

Description:

Sets the attr decl block cursor to the first attr decl block in the current frame and resets the sub-cursors Category and Property

4.3.19 te_nextAttrDecl

int te_nextAttrDecl (PFrameCursor pfc)

Returns:

`true` (non-zero) if there is a next element

Description:

Sets the Property cursor to the next Property class in the attr decl block frame and resets the Category and Property cursors.

4.3.20 te_resetCategory

void te_resetCategory (PFrameCursor pfc)

Description:

Resets the category cursor

4.3.21 **te_nextCategory**

int te_nextCategory (PFrameCursor pfc)

Returns:

`true` (non-zero) if there is a next element

Description:

Sets the category cursor to the next element.

4.3.22 **te_retCategory**

char* te_retCategory (PFrameCursor pfc)

Description:

Returns the category as string

4.3.23 **te_resetProperty**

void te_resetProperty (PFrameCursor pfc)

Description:

Resets the property cursor

4.3.24 **te_nextProperty**

int te_nextProperty (PFrameCursor pfc)

Returns:

`true` (non-zero) if there is a next element

Description:

Sets the category cursor to the next element.

4.3.25 **te_filterPropertyByCategory**

int te_filterPropertyByCategory (PFrameCursor pfc, char* category)

Description:

Lists all properties that are of the type "category". The usage of this function is similar to the function `te_filterPropertyByCategories`. The only difference is the simplicity of the second parameter for the case that you only need to filter with one category.

4.3.26 **te_filterPropertyByCategories**

int te_filterPropertyByCategories (PFrameCursor pfc, char* categories[])

Description:

Lists all properties that matches all types of categories. If the categories are empty then any category matches. In difference to the `nextXXX` functions, these function should be called before the first access via `te_retLabel` or `te_retValue`, because it must search the first valid `AttrDecl`. This means at the beginning you should call: `"te_resetAttrDecl( ... );"` AND `"te_filterPropertyByCategories( ... );"`

4.3.27 te_filterPropertyByLabel

char* te_filterPropertyByLabel (PFrameCursor pfc, char*)

Description:

Lists the value of the property with label of the current frame. This results only one value which is a new created string. Note that the caller has to dispose the return value !

4.3.28 te_retLabel

char* te_retLabel (PFrameCursor pfc)

Description:

Returns the current label of the current property in the current decl block or NULL

4.3.29 te_retValue

char* te_retValue (PFrameCursor pfc)

Description:

Returns the current value of the current property in the current decl block or NULL

Chapter 5

Programming Interface for a Java Client

The Java Application Programming Interface (Java API) consists of a package for the communication with the ConceptBase server, and the Telos Parser which uses the Java Generic Library 3.1.0 of ObjectSpace Inc (<http://www.objectspace.com>). The Telos Parser was generated with the tool JavaCC of SunTest. All classes relevant to ConceptBase are in the packages under `i5.cb`.

This chapter gives only an overview on how to use the Java API of ConceptBase. Detailed documentation of the classes and their methods can be found in the API documentation generated by javadoc. This should be included in the package with programmers information, otherwise contact the ConceptBase Team (cb@i5.informatik.rwth-aachen.de).

The Java API for ConceptBase consists of three main packages:

i5.cb.api contains classes that handle the communication with a ConceptBase, e.g. create a connection, send messages, retrieve answers,

i5.cb.telos.frame contains a Telos parser to parse Telos frames and classes to represent the structure of Telos frames in Java, and

i5.cb.telos.object provides a one-to-one representation of the Telos objects of the ConceptBase server in a Java client. Methods provide facilities to retrieve all instances, subclasses, attributes, etc. of an object.

The preferred method for the interaction with ConceptBase is the usage of the package `i5.cb.telos.object`. The classes of the other packages can be used, too, but then more programming in your client application is required.

Some examples for the communication with ConceptBase can be found in the directory `$CB_HOME/examples/Clients/JavaClient`.

5.1 Communication with ConceptBase: `i5.cb.api`

The main class of the package `i5.cb.api` is the class `CBclient`. The connection with a ConceptBase server can be established during the construction of an object of this class or with the method `enrollMe`.

This class has methods like `tell`, `untell`, `ask` etc. to perform the usual operations on the ConceptBase server. They return in most cases an object of the class `CBanswer`, which represents the answer delivered by the ConceptBase server. The methods and the structures are similar to the methods and structures defined in the C and C++ API.

Furthermore, several `get...`-methods allow to retrieve status information of the client object and some `set`-methods change some parameters of the client, e.g. the timeout value or the current module.

The class `CButil` contains some static methods for decoding and encoding of strings, so that they are accepted by ConceptBase.

The class `CBterm` is used only internally, it parses Prolog-like terms.

Nearly every method throws an exception if some unexpected error has occurred during the operation. All exceptions are derived from the class `i5.cb.CBException`. The exceptions of the class `CBIOException` are thrown if, for example, the communication between client and server is broken or a timeout has occurred. `CBUtilExceptions` are thrown if a string cannot be decoded or encoded.

5.2 Parsing Telos Frames: `i5.cb.telos.frame`

The package `i5.cb.telos.frame` provides all classes and methods that are necessary to parse (and unparse) Telos frames or list of Telos object names, and to represent frames and objects as Java objects.

The parsing of Telos frames or a list of object names requires two steps. First you have to construct a `TelosParser` object:

```
TelosParser tpParser=new
    TelosParser(new StringBufferInputStream(sFrame));
```

The constructor of `TelosParser` requires an `InputStream` object as parameter, therefore it is necessary to construct a `StringBufferInputStream` out of a `String` object.

In the second step, you have to call a method of `TelosParser` to start the parsing. Possible methods are:

telosFrames to parse a set of Telos frames,

telosFrame to parse one Telos frame, and

objectNames to parse a list of Telos object names.

The methods return `TelosFrame(s)` or `ObjectName` objects, that can be accessed with several methods. For details, see the API documentation or the examples provided in `$CB_HOME/examples/Clients/JavaClient`.

It is also possible to construct a `TelosFrame` object step by step, without parsing a string. This is shown in the method `test2` of `ExampleParser.java`. The `TelosFrame` class has a method `toString` which converts the `TelosFrame` into a string, which can be given as an argument to the `tell` method of `CBclient`.

5.3 ObjectBaseInterface: `i5.cb.telos.object`

This package provides methods and classes to represent Telos objects and sets of them as Java objects. There are two possible ways of using this package:

- *without a connection to a ConceptBase server*: Telos objects are created directly in the Java program by using the static methods `getIndividual`, `getSpecialization`, `getInstantiation` and `getAttribute`. Objects may be added to `ITelosObjectSets` that have been created by the `TelosObjectSetFactory`. Within an `ITelosObjectSet`, one can search for instances, subclasses, attributes, etc. of a Telos object.
- *with a connection to a ConceptBase server*: An instance of the class `ObjectBaseInterface` has to be created (using a `CBclient` object). Then, this object can be used to retrieve an object from the ConceptBase server (`getIndividual`), to list all instances of an object (`getAllInstancesOf`), to retrieve all attributes of an object (`getAttributesOf`), etc. The class `ObjectBaseInterface` is an implementation of `ITelosObjectSet`. Also, insertion and deletion of objects in the CB-server via the methods `add` and `remove` is possible. However, it is usually easier to construct a Telos frame and use the method `tell` of `CBclient` than constructing a set of Telos objects.

Note that the relationships of a Telos object to other Telos objects are specific to a Telos object set, e.g. `X` might be an instance of `Y` in one set and not in another set. Therefore, methods such as `getAllInstancesOf` are methods of the `ITelosObjectSet` and not of `TelosObject`.

The file `ExampleOBI.java` in `$CB_HOME/examples/Clients/JavaClient` contains uses the `ObjectBaseInterface` to test various operations.

Appendix A

Example C Client

This chapter explains the usage of the C programming interface with an example program called `LogClient`. This program is able to read the `OB.log` file created by ConceptBase server and performs the operations stored in this file.

The full source code of this program is in the directory `$CB_HOME/examples/Clients/LogClient`. The file `LogClient.c` contains beside the main program some little functions to read the log file. This source file must be compiled and linked together with a version of the ConceptBase library `libCB.a`. The file `MakeLogClient` is a makefile, which executes the necessary commands to compile and link the file with `gcc` on a Unix-platform. The following paragraphs explain only the important parts of the main program.

Before any functions of `libCB` can be used, one must include the header file `CBinterface.h`.

```
#include <CBinterface.h>

int main(int argc, char* argv[]) {

    int PortNr;
    char* HostName;
    char* UserName;
    Answer *ans;
    Server *gserver;
    char* command;
    char* arg;

    char *ClientName = "LogClient";
```

The variables `PortNr`, `HostName`, `UserName` and `ClientName` are initialized with the command line arguments and passed to the `connect_CB_server` function below. `ans` stores the answer of an operation with the ConceptBase server. `gserver` is a pointer to a `Server` structure which is filled by the connect-function. `command` is the command which has been read from the log file and `arg` is the argument for this command.

```
/* Reading and checking command line arguments */
/* ... */

/* Connect to CBserver */
connect_CB_server(PortNr, HostName, ClientName, UserName, &gserver);
if (!gserver) {
    fprintf(stderr, "Connection failed!\n");
    return 1;
}
```


The function `connect_CB_server` opens an IPC socket to the specified ConceptBase server and performs an `ENROLL_ME` method as described in chapter ?? . If the connection can be successfully established, the `gserver` variable points to the connected server. Otherwise, `gserver` will be `NULL`.

Now, the program begins to read the log file. As long as there are commands in the log file, the variable `command` points to a string containing the actual method and `arg` contains the arguments of this method.

Depending on the value of `command`, the program executes the corresponding function to pass the method with its arguments `arg` to the ConceptBase server. Possible values for `command` are e.g. `tellCB`, `untell`, `ask_frames`, ...

```
/* Read commands from logfile until end of file */
while(readLogCommand(fp,&command,&arg)) {

    /* Ask user, if the command should be executed */
    /* ... */

    /* Tell */
    if (!strcmp(command,"tell")) {
        printf("Telling: %s \n\n",arg);
        ans=tellCB( gserver, arg );
    }

    /* Untell */
    if (!strcmp(command,"untell")) {
        printf("Untelling: %s \n\n",arg);
        ans=untell( gserver, arg );
    }

    /* Tell Model */
    if (!strcmp(command,"tell_model")) {
        printf("Loading models: %s \n\n",arg);
        files=commaList2charArray(arg);
        ans=tell_model( gserver, files );
        for(i=0;i<MAX_FILES;i++) {
            if (files[i])
                free(files[i]);
            free(files);
        }
    }
}
```

Note, that the `tell` and `untell` functions take a simple string containing frames as argument, whereas the function `tell_model` takes a list of filenames as argument. The frames are loaded from these files by ConceptBase and the told to the knowledge base.

Tell and untell operations return a pointer to an `Answer` object. For tell and untell, it is sufficient to check the completion value of the answer. The `return_data` can be ignored for these methods.

The following ask functions return also an `Answer` object. The answer of the query is stored in the field `return_data`, the completion is `CB_OK`, if the query could be evaluated. Otherwise the completion will be `CB_ERROR` or `CB_TIMEOUT`.

```
/* Ask objnames */
if (!strcmp(command,"ask_objnames")) {
    printf("Ask (OBJNAMES): %s \n\n",arg);
    ans=ask_objnames( gserver, arg, "LABEL","Now" );
    printf("Answer: %s\n\n", ans->return_data);
}
```

```

/* Ask frames */
if (!strcmp(command, "ask_frames")) {
    printf("Ask (OBJNAMES): %s \n\n", arg);
    ans=ask_frames( gserver, arg, "LABEL", "Now" );
    printf("Answer: %s\n\n", ans->return_data);
}

/* Check completion */
if (ans && ans->completion) {
    fprintf(stderr,
            ">>> Server reports error on method: %s(%s)\n\n",
            command, arg);
}

```

When an error occurred, i.e. completion is not zero (another value than CB_OK), than a error message is printed on the console. Perhaps, it is also useful to get the all error messages from ConceptBase server, but this is not done here¹.

```

}
/* Close connection to CBserver */
disconnect_CB_server(gserver);

return 0;
}

```

If the while-loop is finished the connection to the ConceptBase server can be closed with the function `disconnect_CB_server` and the program is finished.

¹But that should be done in a *good* client program.

Appendix B

Syntax Specifications

B.1 Syntax Specification for IPC messages

<ipcmesssage>	->	ipcmesssage (<sender>,<receiver>,<method_and_args>).
<sender>	->	IPCSTRING
<receiver>	->	IPCSTRING
<method_and_args>	->	<tell> <untell> <ask> <hypoask> <tellmodel> <enrollme> <cancelme> <nextmessage> <stopserver> <reportclients> <lpicall>
<tell>	->	TELL , [<telosframes> <modulearg>]
<tellmodel>	->	TELL_MODEL , [<filelist> <modulearg>]
<filelist>	->	[<ipcstringlist>]
<untell>	->	UNTELL , [<telosframes> <modulearg>]
<ask>	->	ASK , [<askargs> <modulearg>]
<askargs>	->	<query> , <answerrep> , <rollbacktime>
<query>	->	FRAMES , <telosframes> OBJNAMES , <objnames>
<objnames>	->	IPCSTRING
<answerrep>	->	IPCSTRING

<rollbacktime>	->	IPCSTRING
<hypoask>	->	HYPO_ASK , [<telosframes> , <askargs> <modulearg>]
<enrollme>	->	ENROLL_ME , [<toolclass> , <username> <modulearg>]
<modulearg>	->	' , ' IPCID "empty"
<toolclass>	->	IPCSTRING
<username>	->	IPCSTRING
<cancelme>	->	CANCEL_ME , []
<nextmessage>	->	NEXT_MESSAGE , [<method>]
<method>	->	"empty" IPCID
<stopserver>	->	STOP_SERVER , [<method>]
<reportclients>	->	REPORT_CLIENTS , []
<lpicall>	->	LPI_CALL , [IPCSTRING]
<telosframes>	->	IPCSTRING
<ipcstringlist>	->	IPCSTRING <ipcstringlist> , IPCSTRING
IPCSTRING	->	everything enclosed in " except " and \, which must be escape with \
IPCID	->	[a-zA-Z]+[a-zA-Z0-9_]*

B.2 Syntax Specification for IPC answers

<ipcanswer>	->	ipcanswer(<sender>,<completion>,<result>).
<sender>	->	IPCSTRING
<completion>	->	ok error not_handled
<result>	->	IPCSTRING